

SMART DRAIN

Core Objectives

- **Automated Flood Prevention: Real-time urban drainage monitoring to mitigate urban waterlogging.**
- **Smart Maintenance Dispatch: Predictive AI routing for cleanup crews prior to major blockages.**
- **Data Integration: IoT sensor pipeline integrated into city GIS and emergency dispatch networks.**

System Architecture Visual Layout

[IoT Sensors & Flow Meters] → (LoRaWAN Gateway) → [Edge Processing Unit] | [Emergency Services & Mobile App] ◄— (AI Analytics Cloud) ◄┘

Key Milestones & DeliverablesKey Milestones & Deliverables

- **Requirement Specification Document:** Outline power consumption, network range, and sensor accuracy requirements.
- **Component Sourcing Plan:** Choose core hardware components (e.g., NodeMCU/Arduino, Ultrasonic/Float sensors, IP67 weatherproof casing).
- **System Architecture Blueprint:** Map out how field sensors transmit data via cellular/Wi-Fi to cloud databases.

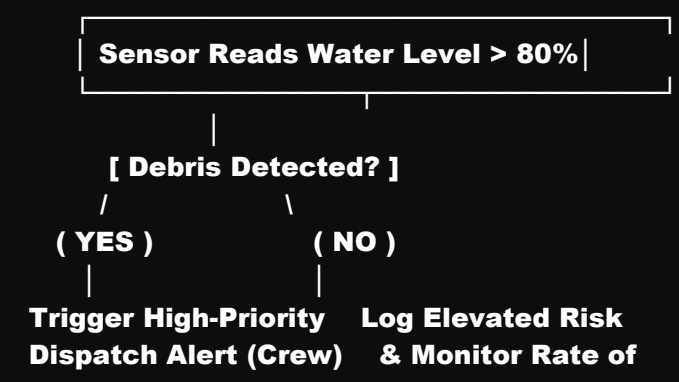
Phase	Duration	Focus Area	Core Deliverables	Visual Infographic Concept
Phase 1: Research & Hardware Design	Months 1–6	Sensor Hardware & Firmware	• Waterproof ultrasonic water-level sensors • Flow rate & debris detection modules • Solar/battery hybrid power system	**Circuit Blueprint Graphic:** Highlighting sensor probes, solar unit, and micro-controller enclosure.
Phase 2: Network & Cloud Platform	Months 7–12	Data Architecture	• LoRaWAN mesh network deployment • Cloud database & real-time telemetry API • Dashboard wireframes for city operators	**Data Pipeline Flowchart:** Showing raw sensor data moving through edge nodes to cloud storage.

Phase	Duration	Focus Area	Core Deliverables	Visual Infographic Concept
Phase 3: AI Analytics & Pilot Test	Months 13–18	Machine Learning Models	• Predictive blockage algorithms • Pilot deployment across 50 urban drain hubs • Field testing under simulated heavy rain	**Split Funnel Graphic:** Inputs (rain gauges, water depth) flowing into AI engine ➔ Outputs (Alerts, Crew Dispatch).
Phase 4: Full Deployment & Integration	Months 19–24	Scaled Rollout	• Deployment across 1,000+ city drainage sites • Integration with municipal GIS platforms • Citizen notification mobile app launch	**City Map Overlay Diagram:** Heatmap showing coverage nodes across downtown, residential, and industrial zones.

Sensor Specifications

- **Detection Type: Ultrasonic depth measurement (Accuracy: $\pm 2\text{ mm}$), optical debris flow monitoring.**
- **Connectivity: LoRaWAN / NB-IoT dual-failover network.**
- **Power Supply: 3.7V LiFePO4 battery pack coupled with a 5V mini solar harvester.**

- **Event Processing Logic:**



Key Performance Indicators (KPIs):

Response Time Reduction:
[ ] 65% Faster
Flood Incident Mitigation:
[ ] 45% Reduction
Maintenance Cost Savings: [

Budget Allocation Breakout:

- **Hardware & R&D: 35%**
- **Software Development & AI: 30%**
- **Pilot & Field Infrastructure: 20%**
- **Operations & Compliance: 15%**

Future Roadmap (Post-Year 2):

- **Autonomous Cleaners: Integration with robotic drain clearing vehicles.**
- **Regional Expansion: Scaling the Smart Drain system to neighboring municipal regions.**